

Persona Name

Sekani Gray

Persona Overview

Sekani Gray is a 23-year-old cybersecurity intern who recently earned a bachelor's degree in Cybersecurity. He has a strong foundation in security concepts such as networking, intrusion detection, and incident response, but limited real-world experience working inside a Security Operations Center (SOC).

As an intern, Sekani wants to build confidence by applying what he learned in school to realistic scenarios. He is motivated to improve his technical skills, better understand security alerts, and prepare for a full-time cybersecurity role.

Demographics

- **Age:** 23
- **Education:** Bachelor's degree in Cybersecurity
- **Job Role:** Cybersecurity Intern
- **Technical Skill Level:** Entry-level to intermediate

Goals

- Practice working with intrusion detection systems
- Learn how to interpret and prioritize security alerts
- Gain hands-on experience responding to cyber threats
- Build confidence before transitioning into a full-time role

Frustrations / Pain Points

- Most training is theoretical and not hands-on

- Real SOC tools can be overwhelming for beginners
- Limited opportunities to safely practice making mistakes
- Difficulty connecting classroom knowledge to real incidents

How Sekani Uses ThreatWatch CyberLab

Sekani utilizes ThreatWatch CyberLab as a hands-on training environment, where he can observe simulated cyberattacks and observe how an intrusion detection system responds. He reviews alerts, decides how to respond, and learns from feedback on his actions. This allows Sekani to practice real-world cybersecurity workflows without the risk of impacting a live system.

Why would Sekani use this app?

Sekani uses ThreatWatch CyberLab because it lets him practice protecting computers from hackers in a safe way. He can see pretend attacks happen, figure out what's wrong, and choose how to stop them. This helps him learn how to do his job better before working on real systems.

Why This Persona Matters

This persona represents an early-career cybersecurity professional who needs practical experience to bridge the gap between education and real-world work. Designing ThreatWatch CyberLab with Sekani in mind ensures the platform focuses on usability, learning, and realistic security workflows rather than unnecessary complexity.

How Sekani Handles a Network Intrusion

When ThreatWatch CyberLab is running in the background, Sekani may receive an alert indicating suspicious network activity, such as multiple connection attempts from a single IP address across different ports. The alert notifies Sekani that a potential network intrusion is occurring and provides contextual information, including the type of activity detected and its severity.

Sekani opens the alert and is presented with a scenario describing the situation in plain, SOC-style language. The app shows relevant indicators such as repeated connection attempts and unusual port access, and offers several response options. These options may include blocking the source IP, restricting access to affected ports, monitoring traffic further, or ignoring the alert.

If Sekani selects an effective response, such as blocking the malicious IP address, the simulated network activity stops, and the system confirms that the intrusion has been contained. If Sekani chooses an ineffective or delayed response, the intrusion escalates, generating additional alerts and requiring further action. Through this loop, Sekani learns how network intrusions are detected, how responses impact system security, and how timely decision-making is critical in real-world security operations.

Persona Name

Christiana Love

Persona Overview

Christiana Love is a 20-year-old undergraduate Computer Science student who is interested in cybersecurity but has limited exposure to security concepts and tools. She has completed introductory programming and networking coursework but has not yet worked with intrusion detection systems, security logs, or incident response processes.

Christiana is curious about cybersecurity as a potential career path, but feels overwhelmed by the complexity of professional security tools. She wants a supportive, low-risk environment where she can learn how cyberattacks are detected and handled without needing advanced prior knowledge.

Demographics

- **Age:** 20
- **Education:** Undergraduate Computer Science major
- **Job Role:** Student (pre-internship)
- **Technical Skill Level:** Beginner

Goals

- Understand how cybersecurity works in real-world settings
- Learn basic intrusion detection and response concepts
- Gain hands-on experience without fear of breaking systems
- Build confidence before taking advanced security courses

Frustrations / Pain Points

- Security terminology and tools feel intimidating
- Learning resources often assume prior security knowledge
- Difficulty visualizing how attacks and defenses interact
- Fear of making mistakes while learning

How Christiana Uses ThreatWatch CyberLab

Christiana uses ThreatWatch CyberLab as an introductory training platform. She begins with guided scenarios that explain alerts in simple language and offer clear response options. Through repeated exposure and feedback, Christiana learns to recognize common attack patterns and understand why certain responses are effective, helping her gradually build foundational cybersecurity skills.

Why would Christiana use this app?

Christiana uses ThreatWatch CyberLab to learn how computers get attacked and how people protect them. The app explains what's happening step by step and lets her practice fixing problems without worrying about messing anything up.

Why This Persona Matters

Christiana represents beginner users who are exploring cybersecurity for the first time. Designing ThreatWatch CyberLab with Christiana in mind ensures the platform remains accessible, educational, and supportive, making cybersecurity concepts easier to understand for users without a strong technical background.

How Christiana Love Handles a Host-Based Intrusion in ThreatWatch CyberLab

While ThreatWatch CyberLab is running in the background, Christiana receives an alert indicating unusual activity on her device, such as an unknown process accessing protected system files. The alert is written in simple language and clearly states that the activity is happening on the device itself, not from the network.

When Christiana opens the alert, the app presents a guided scenario explaining what host-based activity means and why it can be dangerous. The system highlights key indicators such as an unfamiliar process name or unexpected file access, and provides brief explanations to help Christiana understand what she is seeing.

Christiana is then given a small set of response options, such as stopping the suspicious process, reviewing system logs, or ignoring the alert. Each option includes short descriptions to help her make an informed choice. If Christiana selects an appropriate response, such as terminating the suspicious process, the system shows that the abnormal activity stops and confirms that the threat has been contained. If she selects an ineffective option, the intrusion escalates, and additional alerts appear, prompting further guidance and learning.

Persona Name

Dr. Angel Ryder

Persona Overview

Dr. Angel Ryder is a cybersecurity instructor and professor who teaches undergraduate and early graduate-level security courses. She has a strong background in cybersecurity concepts and professional experience, but is primarily focused on helping students translate theory into real-world practice.

Dr. Ryder wants a safe, structured platform that allows students to experience intrusion detection and response workflows without the cost, complexity, or risk of enterprise security systems.

Demographics

- **Age:** 47
- **Education:** Master's in Cybersecurity
- **Job Role:** Cybersecurity Instructor / Trainer
- **Technical Skill Level:** Advanced

Goals

- Provide students with hands-on cybersecurity experience
- Demonstrate how intrusion detection systems work in practice
- Teach decision-making and response workflows, not just theory
- Evaluate student understanding through realistic scenarios

Frustrations / Pain Points

- Limited access to realistic SOC-style environments for teaching
- Enterprise tools are too complex or expensive for classroom use
- Traditional labs focus on exploitation rather than defense
- Difficulty assessing how well students understand response processes

How Dr. Angel Ryder Uses ThreatWatch CyberLab

Dr. Ryder uses ThreatWatch CyberLab as an instructional and training tool. She assigns scenarios to students, allows them to interact with simulated intrusions, and reviews performance summaries to assess how students interpret alerts and respond to incidents. The platform supports guided learning while allowing Dr. Ryder to scale hands-on practice across multiple students.

Why would Dr. Ryder use this app?

Dr. Ryder uses ThreatWatch CyberLab to help her students practice protecting computers from hackers in a safe way, so they can learn by doing instead of just reading about it.

Why This Persona Matters

Dr. Ryder represents instructors and trainers who shape how cybersecurity is taught. Designing ThreatWatch CyberLab with her needs in mind ensures the platform supports structured learning, guided progression, and assessment, making it suitable for classroom and training environments.

How Dr. Ryder Handles a Password Cracking Intrusion

When a password cracking intrusion occurs in ThreatWatch CyberLab, the intrusion detection system identifies abnormal authentication behavior, such as repeated failed login attempts across multiple accounts from the same source. The system generates an alert that is delivered to students using the platform, while Dr. Ryder has visibility into the scenario through an instructor view.

Dr. Ryder uses this intrusion scenario as a teaching and assessment opportunity. She allows students to receive the alert, analyze the authentication activity, and select response actions such as locking accounts or blocking the source IP. As students interact with the scenario, the platform records their decisions, timing, and outcomes.

If students choose effective responses, the system shows that login attempts stop and the intrusion is contained. If students make ineffective choices, the intrusion escalates, allowing Dr. Ryder to observe how students adapt and learn from feedback. After the scenario concludes, Dr. Ryder reviews performance summaries to evaluate how well students understood authentication-based attacks and defensive response strategies.

Through this process, Dr. Ryder uses ThreatWatch CyberLab to demonstrate how password cracking attacks are detected, how response decisions impact system security, and how students develop practical incident response skills in a controlled environment.

Network Intrusion

What it is

A network intrusion occurs when an unauthorized user gains access to a network by exploiting weaknesses in network configurations, services, or security controls.

Example

An attacker scans a company's public IP range, discovers an exposed SSH service, and gains access to the internal network using weak or reused credentials.

Common methods

- ❖ Port scanning to identify open or exposed services
- ❖ Exploiting misconfigured firewalls or routers
- ❖ Accessing unsecured remote services (SSH)
- ❖ Man-in-the-Middle attacks on unsecured networks

How to prevent it

- ❖ Proper firewall configuration and network segmentation
- ❖ Closing unused ports and services
- ❖ Strong authentication for remote access (VPN)
- ❖ Continuous network monitoring and intrusion detection systems (IDS)

How the IDS sees it:

- ❖ Multiple connection attempts from the same IP address
- ❖ Rapid access to different ports within a short time window
- ❖ Connection attempts to restricted or unused services
- ❖ Unusual traffic volume compared to normal behavior

What triggers an alert:

- ❖ Exceeding a defined threshold of connection attempts
- ❖ Access patterns consistent with scanning or probing behavior

What the IDS does:

- ❖ Detects abnormal network behavior

- ❖ Generates alerts describing suspicious traffic patterns
- ❖ Tracks whether the activity continues or escalates

How resolution happens in the app:

- ❖ The user is presented with response options such as:
 - Block the source IP
 - Restrict access to specific ports
 - Monitor traffic further
- ❖ If the correct action is taken:
 - The simulated malicious traffic stops
 - The IDS confirms the threat is contained
- ❖ If the wrong action is taken:
 - Network activity escalates
 - Additional alerts are generated

Host-Based Intrusion

What it is

A host-based intrusion involves compromising an individual system, such as a server or workstation, to gain control over the operating system or installed applications.

Example

A user opens a malicious email attachment, allowing an attacker to install malware that grants administrative access to the user's computer.

Common methods

- ❖ Malware or Trojan installation
- ❖ Exploiting unpatched operating systems or software vulnerabilities
- ❖ Privilege escalation from a standard user to an administrator
- ❖ Installing backdoors or persistence mechanisms

How to prevent it

- ❖ Regular system updates and patch management
- ❖ Endpoint protection software (antivirus)
- ❖ Least-privilege access controls
- ❖ Host-based intrusion detection systems (HIDS)

How the IDS sees it:

- ❖ Unexpected access to protected system files
- ❖ Execution of unfamiliar or unauthorized processes
- ❖ Privilege-related actions outside normal usage patterns
- ❖ Changes to system configurations without authorization

What triggers an alert:

- ❖ An activity that violates predefined host-level security rules
- ❖ Actions that deviate from established behavioral baselines

What the IDS does:

- ❖ Detects suspicious system-level activity

- ❖ Alerts the user to unauthorized file or process behavior
- ❖ Monitors continued host activity after detection

How resolution happens in the app:

- ❖ The user chooses actions such as:
 - Terminating a suspicious process
 - Restricting file access
 - Investigating system logs
- ❖ Correct actions:
 - Stop further suspicious activity
 - Return host status to normal
- ❖ Incorrect actions:
 - Allow persistence or privilege escalation
 - Trigger higher-severity alerts

Password Cracking

What it is

Password cracking is the process of attempting to recover or guess passwords to gain unauthorized access to systems, applications, or networks.

Example

An attacker uses a database of leaked usernames and passwords to repeatedly attempt VPN logins until a valid credential is found.

Common methods

- ❖ Brute-force attacks that try all possible password combinations
- ❖ Dictionary attacks using common or leaked passwords
- ❖ Credential stuffing using credentials from previous data breaches
- ❖ Offline cracking of stolen password hashes

How the IDS sees it:

- ❖ Multiple failed login attempts within a short time frame
- ❖ Login attempts across multiple user accounts
- ❖ Repeated authentication attempts from the same IP address
- ❖ Abnormal login timing or frequency

How to prevent it

- ❖ Strong password policies (length, complexity, uniqueness)
- ❖ Multi-factor authentication (MFA)
- ❖ Account lockout policies after repeated failed attempts
- ❖ Secure password storage using salted hashing

What triggers an alert:

- ❖ Authentication failures exceeding configured thresholds
- ❖ Patterns consistent with brute-force or credential-stuffing attacks

What the IDS does:

- ❖ Detects abnormal authentication behavior

- ❖ Raises alerts for repeated failed login attempts
- ❖ Tracks whether login attempts continue

How resolution happens in the app:

- ❖ The user may:
 - Lock affected accounts
 - Block the source IP
 - Enforce stronger authentication controls
- ❖ Correct responses:
 - Stop further login attempts
 - Mark the intrusion as contained
- ❖ Incorrect responses:
 - Allow eventual account compromise
 - Escalate the scenario